```
      "location": {
        "lat": 50.00,
        "lon": 0.10
      }
    }
  }
}
```

Sets the central location, defined as a point on the map

As the listing shows, the `geo_distance` query expects two attributes: `distance`, which provides the radius of the geocircle, and `location`, which defines the central point of the geocircle. The query returns all restaurants within 175 km of the defined point.

> **NOTE** The `distance` field accepts the distance measured in kilometers or miles as `km` or `mi`, respectively. Elasticsearch also honors a value given as "`350 mi`" as well as `"350mi"` (with the space removed).

It shouldn't surprise you that the field for defining the central point can be input using the latitude and longitude in the form of strings, arrays, WKT, and other formats. Queries can also be run on geoshapes, although I leave it to you to experiment with them.

The `geo_bounding_box` and `geo_distance` queries let us search for addresses represented as point-based locations in rectangular shapes and circles. However, we often need to search for an address defined as a shape inside another shape, preferably a polygon. This is where we use a `geo_shape` query.

## 12.6 The geo_shape query

Not all locations are point-based (coordinates with latitude and longitude values). Sometimes we want to determine whether shapes are present inside (or outside) the boundaries of another shape or whether they intersect the boundaries. For example, figure 12.6 shows some plots of land on our London map.

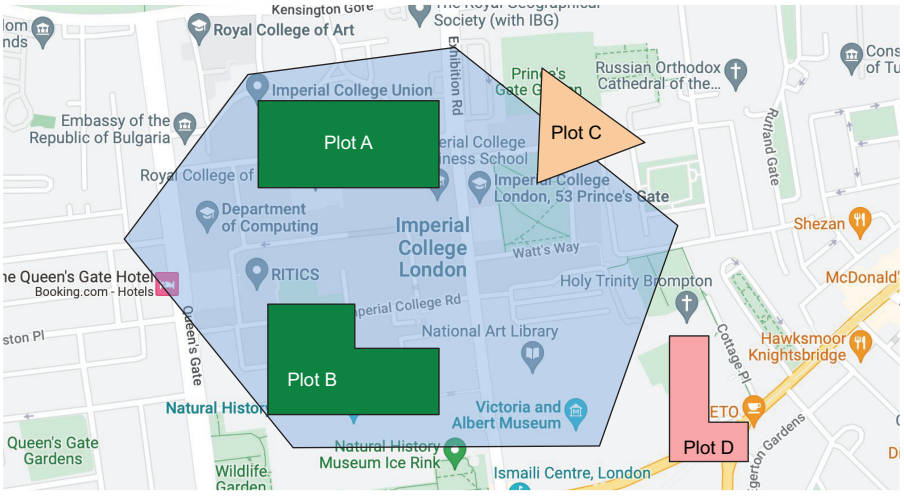



**Figure 12.6 Plots of farmland in an area of London**